

Contact

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ganeshreddy09 (LinkedIn)

[gsr07.github.io/Portfolio-Ganesh/](https://github.com/gsr07)
(Portfolio)

Top Skills

Platform Io

Open CV

Motion Planning

Languages

German (Elementary)

Certifications

ROS2 certificate

Ganesh Reddy

Master's Student in Autonomous Systems | ROS2, C++, Python |
Robotics | Machine Learning
Erlangen, Bavaria, Germany

Summary

Passionate Master's student in Autonomy Technologies at FAU Erlangen with hands-on experience in robotics, embedded systems, and computer vision. I specialize in bridging hardware and software to design intelligent robotic systems that are reliable, efficient, and industry-ready.

Skilled in ROS2, Nav2, Python, C++, Embedded C, and MATLAB, with strong expertise in navigation, perception, control systems, and real-time algorithm development. My background combines CAD design (SolidWorks), PCB design (KiCad, Eagle), and embedded programming (ESP32, Arduino, Raspberry Pi) with advanced computer vision techniques (OpenCV, YOLO, PyTorch, TensorFlow).

Key experiences include:

Developing an Autonomous Mobile Robot (AMR) with 300kg payload capacity, integrating EKF-based localization, A navigation, and robotic arm kinematics*.

Building IoT systems with ESP32, BLE-based localization, and custom PCBs for industrial automation.

Leading computer vision projects such as real-time object detection (YOLO), driver drowsiness detection, and medical imaging (lung disease classification).

Designing and simulating autonomous robots, robotic arms, drones, and line-following systems in ROS/Gazebo/Simulink.

I thrive at the intersection of robotics, embedded systems, and AI-driven perception, aiming to contribute to innovative solutions in mechatronics, automation, and autonomous systems.

Let's connect if you're looking for a problem-solver who can design, prototype, and integrate intelligent systems from PCB to perception.

Experience

Department Artificial Intelligence in Biomedical Engineering -
AIBE@FAU

Student Teaching Assistant

November 2025 - Present (4 months)

Erlangen

Digital Associates India

Internet of Things Engineer

January 2023 - October 2023 (10 months)

Hyderabad, Telangana, India

Developed BLE-based indoor localization systems and IoT dashboards using ESP32, custom PCBs, and environmental sensors (temperature, humidity, fire). Built firmware using ESP-IDF (C, Python) for real-time sensor communication. Contributed to technical documentation, Gantt-based project tracking, and Agile ceremonies as part of a cross-functional team.

Centre for autonomous systems engineering

System Engineering Intern

January 2022 - July 2022 (7 months)

Hubli, Karnataka, India

Designed and fabricated a 5DOF robotic arm and AGV components using SolidWorks, NX-CAD, and ROS-Gazebo, integrating motion planning and kinematics. Tuned Dynamixel XM motors and implemented control architecture for full system functionality in simulation and real-world testing.

Education

FAU Erlangen-Nürnberg

Master's degree, Autonomy Technologies · (April 2024 - September 2026)

KLE Technological University - Hubballi (India)

Bachelor of Engineering - BE, Mechatronics, Robotics, and Automation Engineering · (2018 - 2022)